

Sri Lanka Institute of Information Technology

IT3060 – Human Computer Interaction

BSc (Hons) in IT 3rd Year 2nd Semester

Faculty of Computing – SLIIT

Lab Exercise 03 Report



FitFlow Redesign

Design, Wireframes & Prototype – Lab Exercise 03 Report

Student Name	MARASINGHE M M W K
IT No	IT23606760
Batch	Y3.S2.IT_WE_02.02
Case Study	FitFlow Redesign Scenario

Contents

Case Study Overview – FitFlow Redesign	3
Activity 1: Review User Requirements	4
Activity 2: Perform Crazy 8 Sketching.....	6
Activity 3: Produce Multiple Interface Variants.....	7
Activity 4: Select the Best Design (Decision Matrix & Justification).....	8
Activity 5: Create Paper & Digital Wireframes.....	9
Wireframe 1 — Home Dashboard	9
Wireframe 2 — AI Workout Planner.....	10
Wireframe 3 — Progress Tracking	11
Wireframe 4 — Community Feed	12
Wireframe 5 — Nutrition Logger	13
Activity 6: Build a Clickable Low-Fidelity Prototype.....	14

Case Study Overview – FitFlow Redesign

FitFlow is a fitness-tracking app from a mid-sized health-tech startup that had been steadily losing users, with app-store ratings falling from 4.6 to 3.8 stars and 68% of new users abandoning the app shortly after onboarding. Lab 01 planned and conducted the user research; Lab 02 analysed it into personas, journeys, How-Might-We statements and a prioritised requirements set. This report (Lab 03) turns that understanding into design: reviewing requirements, sketching ideas with Crazy 8s, producing and evaluating interface variants, and building low-fidelity wireframes and a clickable prototype.

The three redesign priorities carried through every activity are AI-powered personalised workouts, social community features, and improved nutrition tracking, mapped to five core screens: Home Dashboard, AI Workout Planner, Progress Tracking, Community Feed and Nutrition Logger.

Activity 1: Review User Requirements

The research findings and pain points from Labs 01–02 were reviewed and translated into a prioritised requirements list and a set of core user flows that drive the rest of the design.

Pain Points

- Poor progress visualisation — no rewards or celebratory feedback, so motivation fades.
- Cumbersome logging — manual workout and nutrition entry is tedious and abandoned after a few days.
- Limited personalisation — generic plans that ignore goals and schedule (the strongest theme).
- Supporting issues — social isolation and onboarding friction (68% early drop-off).

Key User Needs (New Features)

- AI-powered adaptive workout plans (personalisation + flexibility).
- A private social community for accountability and motivation.
- Fast, camera-based nutrition tracking to remove logging friction.
- Clear progress dashboards with rewards for visualisation and motivation.
- A streamlined onboarding flow to cut early drop-off.

Prioritised Requirements

ID	Requirement	Priority	Core Flow	Source
R1	AI generates adaptive plans from user goals and available time	High	Workout Planner	Alex / Personalisation
R2	Beginner-friendly AI coaching with safe exercise suggestions	High	Workout Planner	Priya / Personalisation
R3	Short, guided onboarding leading to a first workout	High	Onboarding → Home	Priya / Onboarding
R4	Camera-based nutrition logging powered by computer vision	High	Nutrition Logger	Sam / Nutrition
R5	Progress dashboard with charts, streaks and celebrations	High	Progress Tracking	All / Motivation
R6	AI “Daily Flow” recommendation card on the home screen	High	Home Dashboard	Alex / Personalisation
R7	Private social circles with challenges and group interactions	Medium	Community Feed	Priya, Alex / Social
R8	Achievement badges and streaks for consistency	Medium	Progress / Home	All / Motivation
R9	Smart, personalised workout reminders	Medium	Home / Workout	Alex / Flexibility
R10	Share achievements to the community feed	Low	Community Feed	Sam / Social

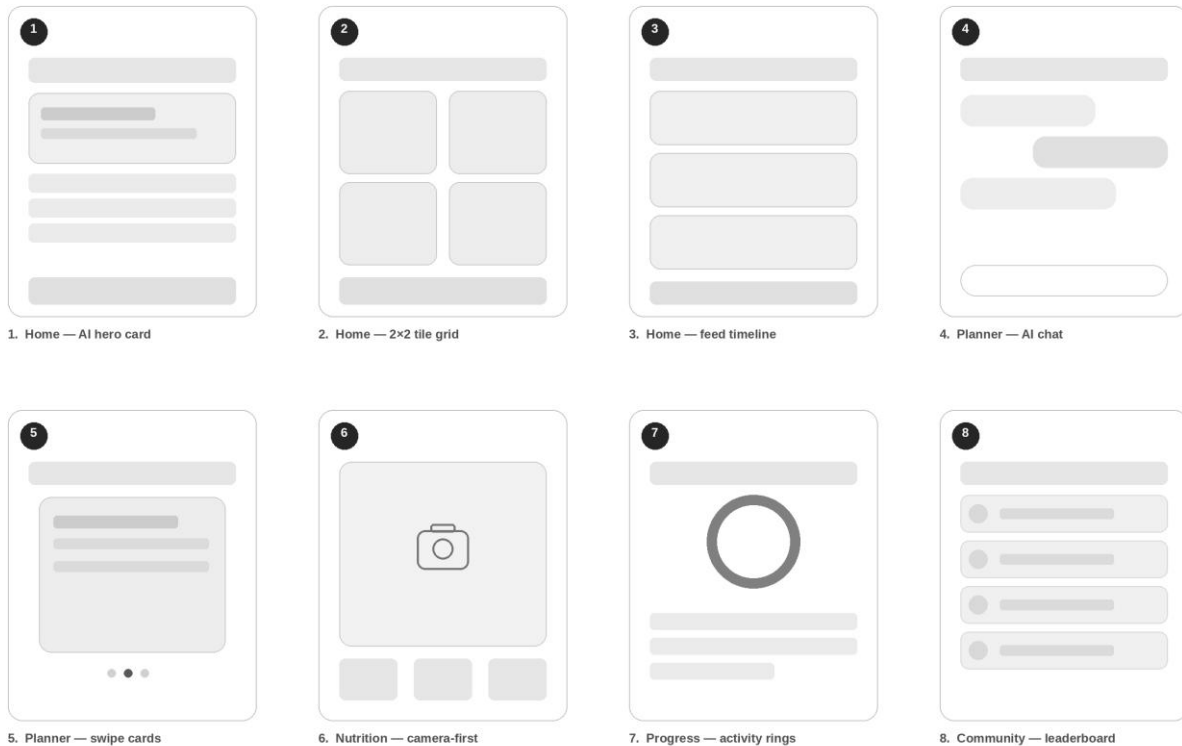
Core User Flows

- Home Dashboard — launch → AI “Daily Flow” + progress snapshot → start workout, log a meal, or open community.

- AI Workout Planner — confirm goals + time → AI generates plan → review / tweak → start and log the session.
- Progress Tracking — open dashboard → workout / nutrition / streak charts → badge on milestones → history.
- Community Feed — open feed → browse challenges / groups in a private circle → join → post / react.
- Nutrition Logger — tap camera → AI recognises food + estimates calories → confirm / adjust → saved to log.

Activity 2: Perform Crazy 8 Sketching

Using the Crazy 8s method, eight interface ideas were sketched rapidly (eight sketches in eight minutes) across the core screens — Home Dashboard, AI Workout Planner and additional Progress, Nutrition and Community concepts. The goal was quantity and range of ideas rather than polish, giving a broad pool to draw the interface variants from.



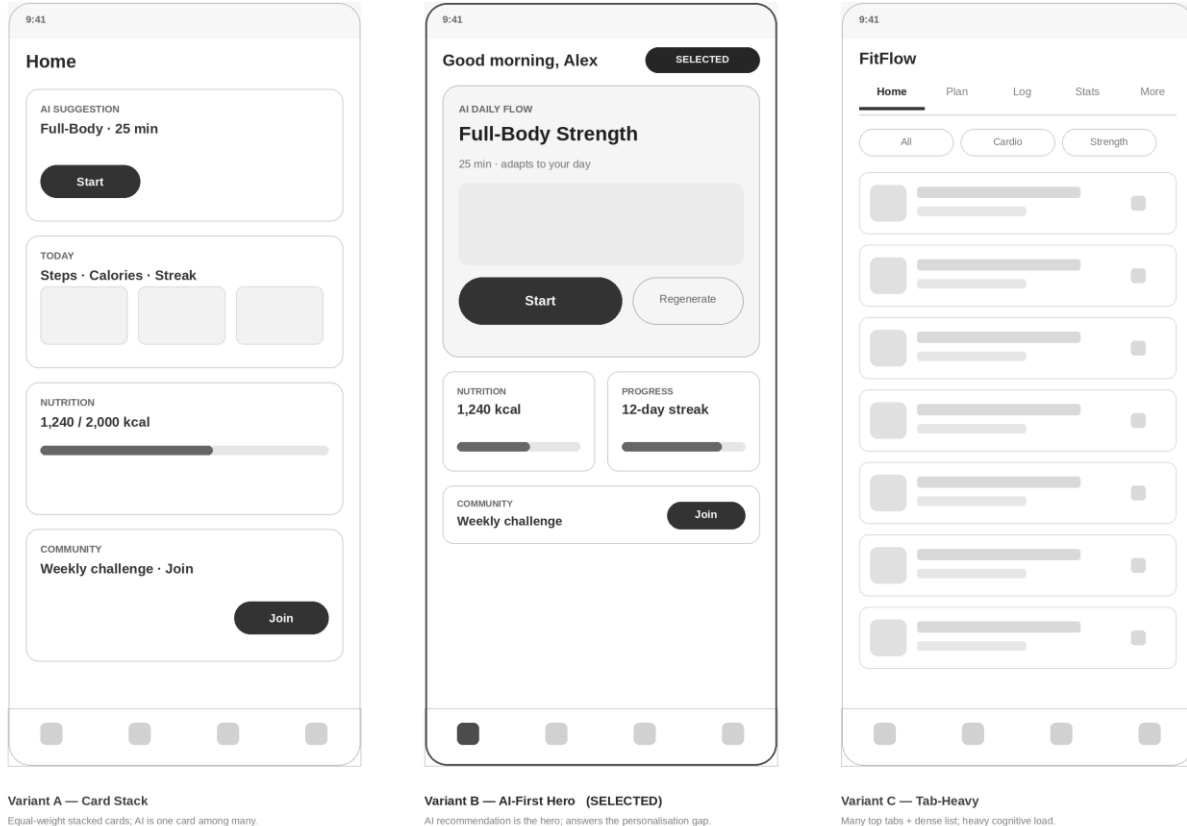
Crazy 8 sketches — eight rapid interface ideas for FitFlow

The Eight Ideas

- Home — AI hero card: a dominant AI suggestion at the top with a supporting list.
- Home — 2x2 tile grid: equal tiles for workout, nutrition, progress and community.
- Home — feed timeline: a vertical, scrollable feed of cards.
- Planner — AI chat: a conversational interface that builds the plan through dialogue.
- Planner — swipe cards: one exercise per card, swiped left/right to accept or swap.
- Nutrition — camera-first: a full-screen capture area as the primary action.
- Progress — activity rings: a large ring visual summarising daily goals.
- Community — leaderboard: a ranked list to drive friendly competition.

Activity 3: Produce Multiple Interface Variants

Drawing on the Crazy 8s, three distinct interface variants of the Home Dashboard were developed. Each takes a different approach to layout, navigation, visual hierarchy and how AI recommendations and social elements are surfaced.



Three interface variants of the Home Dashboard

- Variant A — Card Stack: equal-weight stacked cards. Familiar and safe, but AI is just one card among many, so it under-sells the personalisation the research demands.
- Variant B — AI-First Hero (selected): the AI recommendation is the hero element, with compact secondary tiles and a slim community strip below. Directly foregrounds personalisation.
- Variant C — Tab-Heavy: a dense top-tab bar over a long list. Powerful for advanced users but deep and cognitively heavy, weakest for the beginners in the personas.

Activity 4: Select the Best Design (Decision Matrix & Justification)

The three variants were scored (1 = weak, 5 = strong) against the criteria set by the lab: usability, visual appeal, alignment with user needs, innovation and feasibility.

Criterion	Variant A — Card Stack	Variant B — AI-First Hero	Variant C — Tab-Heavy
Usability	4	5	3
Visual appeal	4	4	3
Alignment with user needs	4	5	3
Innovation	3	5	2
Feasibility	5	4	4
Total	20	23	15

Selected direction: Variant B — AI-First Hero.

Justification

Variant B was selected as the primary design direction because it scored highest overall (23/25) and, more importantly, it directly targets the strongest theme from the research: the personalisation gap. Across Lab 01 and Lab 02, “generic plans” was the most frequently cited pain point, appearing in interviews, observations and app-store reviews. Variant B answers this by making the AI “Daily Flow” recommendation the hero element of the Home Dashboard — the first thing a user like Alex Rivera sees is a workout already adapted to his available time, rather than a generic plan he has to hunt for. This reframes FitFlow from a passive tracker into an active coach, which is the core promise of the redesign.

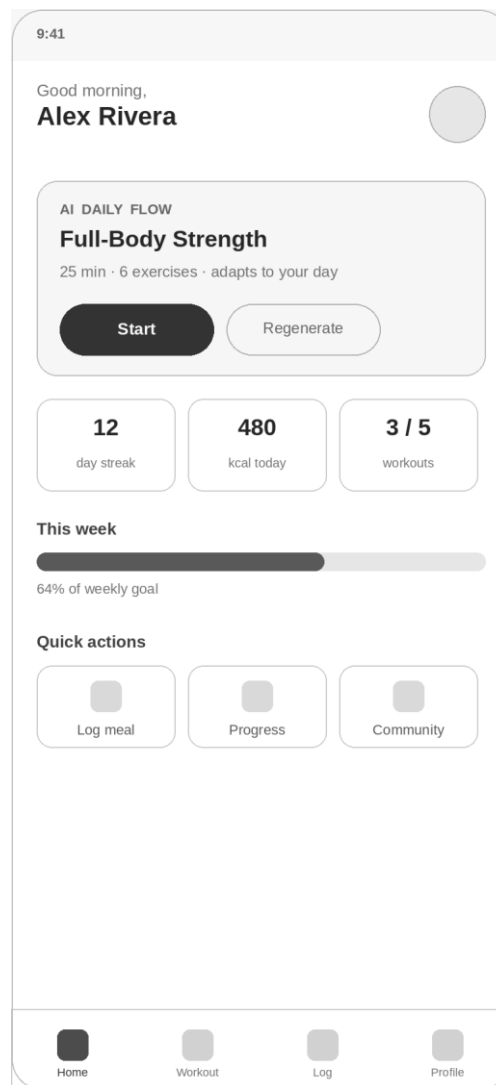
The design also addresses the two other validated themes. Tracking friction is reduced through a camera-first Nutrition Logger, where a single large capture button replaces tedious manual entry — the exact behaviour Sam Fernando abandoned “after a few days.” Waning motivation and social isolation are handled by surfacing streaks, badges and a private-circle Community feed, giving Priya Singh the accountability and encouragement she needs without an overwhelming public network.

On usability, Variant B keeps a shallow, three-tap navigation with a persistent bottom bar, which usability heuristics favour over the deep tab hierarchy of Variant C (scored lowest at 15). Variant A (Card Stack) was a close, safer second at 20 and remains a strong fallback, but it treats AI as one card among many rather than the centrepiece, so it under-delivers on the personalisation theme that matters most. Feasibility remains realistic: the layout maps cleanly onto the case study’s chosen stack (React Native front end, TensorFlow Lite on-device personalisation, ML-kit computer vision for food recognition), so nothing in the design outruns what the team committed to build. For these reasons — highest alignment with the top user need, strong innovation and buildable within scope — Variant B is taken forward into the wireframes and prototype.

Activity 5: Create Paper & Digital Wireframes

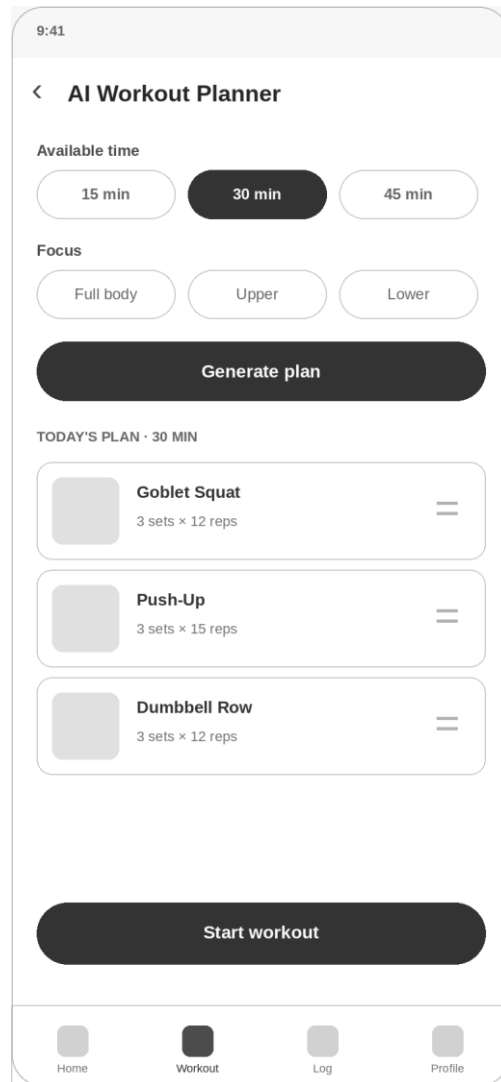
Low-fidelity ideas were first drafted as quick hand sketches (the Crazy 8s above serving as the rapid paper ideation), then the strongest concept — Variant B — was converted into digital low-fidelity wireframes in Figma for all five core user flows. Following the deliverable specification, the wireframes use grayscale and basic shapes only, with no colour or detailed icons, and focus on layout, user flow, microcopy and interaction patterns.

Wireframe 1 — Home Dashboard



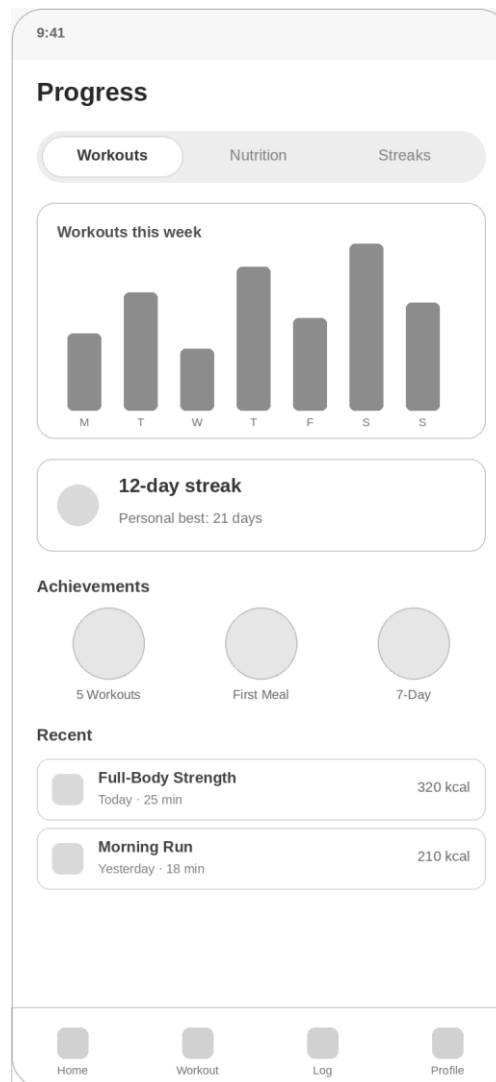
The AI “Daily Flow” card is the visual hero, backed by streak / calorie / workout stats, a weekly-progress bar and one-tap quick actions. Primary flow: launch → start the suggested workout, log a meal, or open the community.

Wireframe 2 — AI Workout Planner



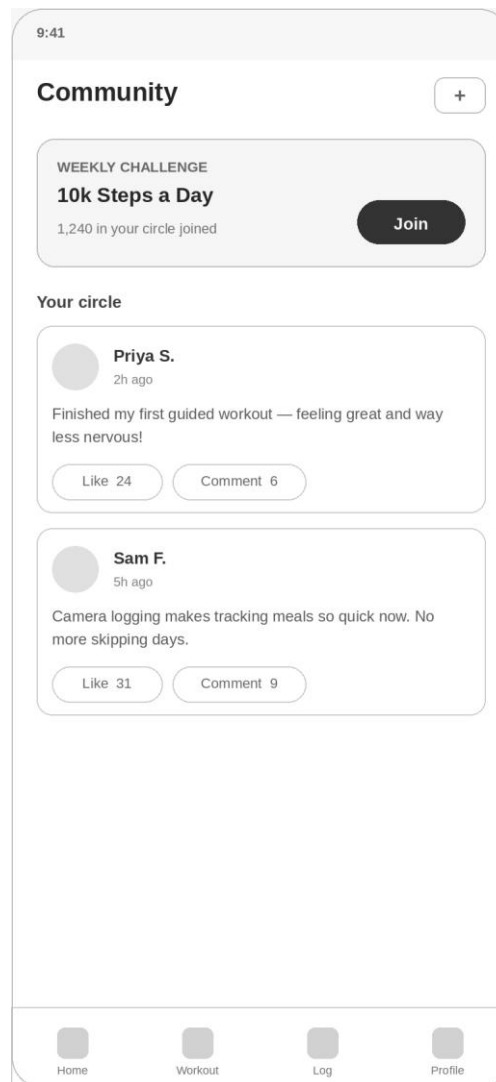
Users pick available time and focus, tap Generate, then review an adaptive exercise list with drag handles before starting. Flow: set time → generate → tweak → start workout.

Wireframe 3 — Progress Tracking



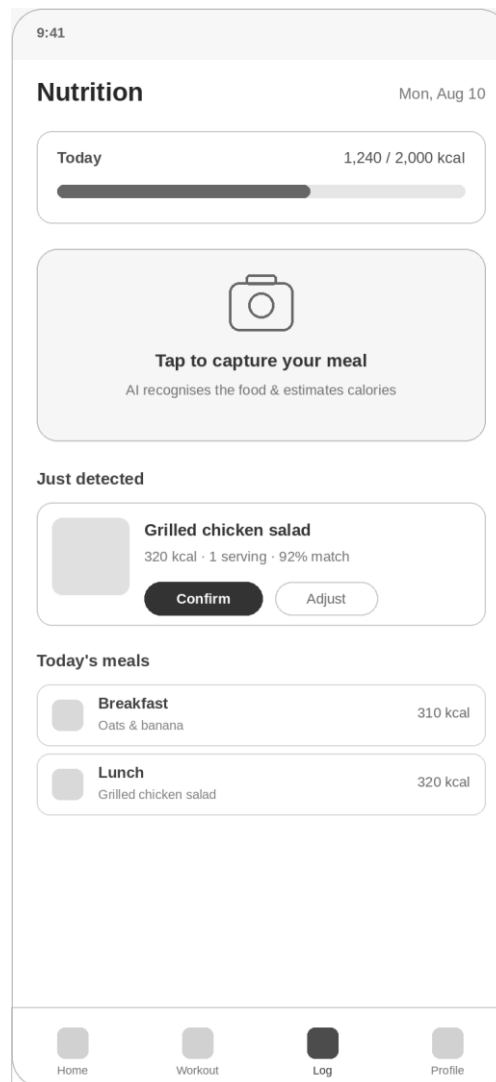
Segmented tabs switch between workouts, nutrition and streaks; a bar chart, streak card, achievement badges and recent history give celebratory feedback that addresses waning motivation.

Wireframe 4 — Community Feed



A weekly-challenge banner with a clear Join action sits above private-circle posts with like / comment controls — social support without an overwhelming public network.

Wireframe 5 — Nutrition Logger



A single large camera-capture button is the primary action; the AI-detected item is shown with Confirm / Adjust, feeding a simple daily meal list. This removes the manual-logging friction users abandoned.

Activity 6: Build a Clickable Low-Fidelity Prototype

The five digital wireframes were linked into a clickable low-fidelity prototype in Figma. All major user flows are connected to demonstrate navigation and basic interactions, with the Home Dashboard set as the starting point.

Prototype Link

<https://www.figma.com/design/CIQq1FIYMX2kqXM0hUJXF>
[HCI-Lab-03-Wireframes Prototype Link](#)

Open the file and press Present to launch the prototype from the Home Dashboard; every wired button and navigation item is clickable.

Wired Flows

- Persistent bottom navigation on every screen: Home ↔ Workout ↔ Log (Nutrition).
- AI “Daily Flow” Start button → AI Workout Planner.
- Home quick actions → Nutrition, Progress and Community.
- “Start workout” → Progress Tracking, closing the loop with streaks and badges.

Key Design Decisions (Annotations)

- The AI recommendation is the hero of Home, directly answering the personalisation gap (top research theme).
- Quick actions cut navigation friction to a single tap.
- “Start workout” → Progress provides celebratory feedback for waning motivation.
- The Nutrition Logger is camera-first, replacing tedious manual entry.
- Fidelity: grayscale, basic shapes only, no colour or detailed icons — per the Activity 5 spec.